

Homotopy.

Def. Let  $f, g: X \rightarrow Y$  be continuous maps.

We said that  $f$  and  $g$  are Homotopic if there exists a Continuous map

$$F: X \times I \rightarrow Y \text{ such that } \begin{cases} F(x, 0) = f(x), & \forall x \in X. \\ F(x, 1) = g(x), & \forall x \in X. \end{cases}$$

This  $F$  is called a Homotopy, and in such a case, denote  $f \simeq g$ .

Thm. Let  $C(X, Y) \stackrel{\text{def}}{=} \{f: X \rightarrow Y \mid f \text{ is continuous}\}$ ,

Define a relation on  $C(X, Y)$  as:  $f \sim g \Leftrightarrow f \simeq g$ .

Then, the Homotopy is equivalence relation.

Proof. Reflexivity: Given  $f \in C(X, Y)$ , define

$$F: X \times I \rightarrow Y: (x, t) \mapsto f(x)$$

Then, since  $F^{-1}[V] = f^{-1}[V] \times I$ ,  $F$  is continuous such that  $F(x, 0) = F(x, 1) = f(x)$ .

Symmetric. Let  $f \simeq g$  via  $F: X \times I \rightarrow Y$ . Define  $G: X \times I \rightarrow Y: (x, t) \mapsto F(x, 1-t)$

Then,  $G(x, 0) = F(x, 1) = g(x)$  and  $G(x, 1) = F(x, 0) = f(x)$ .

Transitivity. Let  $f \simeq g$  via  $F$  and  $g \simeq h$  via  $G$ . Define

$$H: X \times I \rightarrow Y: (x, t) \mapsto \begin{cases} F(x, 2t) & \text{if } 0 \leq t \leq \frac{1}{2} \\ G(x, 2t-1) & \text{if } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Then, since  $[0, \frac{1}{2}]$  and  $[\frac{1}{2}, 1]$  are closed with  $F(x, 1) = g(x) = G(x, 0)$ , therefore, the Pasting lemma gives that the  $H$  is Homotopy from  $f$  to  $h$ . So  $f \simeq h$ .

Def. Let  $X$  and  $Y$  be topological space.

We said that  $X$  and  $Y$  are Homotopy equivalent if there exists two Continuous map  $f: X \rightarrow Y$  and  $g: Y \rightarrow X$  such that  $g \circ f \simeq \text{id}_X$  and  $f \circ g \simeq \text{id}_Y$ .

In such a case, denote  $X \simeq Y$ .

And, such a  $f: X \rightarrow Y$  is called a Homotopy equivalence, and such a  $g: Y \rightarrow X$  is called a Homotopy inverse.

Note: Trivially, Homeomorphism is Homotopy equivalence. Therefore,

$X \cong Y \Rightarrow X \simeq Y$ , but the Converse does not hold.

Counterexample: Consider two subspaces  $S', \mathbb{R}^2 \setminus \{0\} \subseteq \mathbb{R}^2$ .

First, since  $S'$  is closed and bounded, so it is Compact.

And, since  $\{0\}$  is closed,  $\mathbb{R}^2 \setminus \{0\}$  is open, so it is not Compact. So,  $S' \not\simeq \mathbb{R}^2 \setminus \{0\}$ .

Next, to show that  $S' \simeq \mathbb{R}^2 \setminus \{0\}$ , Consider that an inclusion

$$f: S' \hookrightarrow \mathbb{R}^2 \setminus \{0\}; x \mapsto x \quad \text{and}$$

$$g: \mathbb{R}^2 \setminus \{0\} \rightarrow S'; x \mapsto \frac{x}{\|x\|}.$$

Then, both  $f$  and  $g$  are Continuous, and  $g \circ f = \text{id}_{S'}$ .

And,  $(f \circ g): \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\}; x \mapsto x/\|x\|$  is Homotopic to  $\text{id}_{\mathbb{R}^2 \setminus \{0\}}$  via

$$F: [\mathbb{R}^2 \setminus \{0\}] \times [0, 1] \rightarrow \mathbb{R}^2 \setminus \{0\}; (x, t) \mapsto (1-t) \cdot \frac{x}{\|x\|} + t \cdot x.$$

which satisfies  $F(x, 0) = x/\|x\|$  and  $F(x, 1) = x$ , so Prove.

Contractible

Def.

A space that homotopy equivalent to a point is called Contractible.

Thm. Let  $X$  be a Topological space. Then, TFAE:

a).  $X$  is Contractible. (Homotopy equivalent to a point)

b).  $\text{id}_X: X \rightarrow X: x \mapsto x$  is null-homotopic.

c). Given  $x \in X$ ,  $C_x \simeq \text{id}_X$  where  $C_x: X \rightarrow X: z \mapsto x$ .

Proof.

Assume a). That is, there exist two maps

$f: X \rightarrow \{p\}$  and  $g: \{p\} \rightarrow X$  such that  $f \circ g \simeq \text{id}_p$  and  $g \circ f \simeq \text{id}_X$ .

Then, since  $f: X \rightarrow \{p\}$  must be a Constant Mapping  $x \mapsto p$ ,

$g \circ f(x) = g(p)$  for all  $x \in X$ , so  $g \circ f$  is Constant, so  $\text{id}_X$  is null-homotopic.

It proved b).

Assume b). Let  $C: X \rightarrow X: z \mapsto x$  be a Constant map such that  $C \simeq \text{id}_X$ , via

$H: X \times I \rightarrow X$  such that  $H(z, 0) = z$  and  $H(z, 1) = x$ .

Let  $x \in X$  be fixed. Define a path  $p: [0, 1] \rightarrow X: t \mapsto H(x, t)$ .

Then,  $p(0) = H(x, 0) = x$  and  $p(1) = H(x, 1) = x$ .

Now,  $F: X \times I \rightarrow X: (z, t) \mapsto p(t)$  is the Homotopy to  $C \simeq C_x$ . Now, by transitivity,  $C \simeq \text{id}_X$  and  $C \simeq C_x$  implies that  $C_x \simeq \text{id}_X$ . It proved c).

Assume c). Let  $f: X \rightarrow \{p\}: x \mapsto p$  be a map and let  $x \in X$  be given

Then, a map  $g: \{p\} \rightarrow X: p \mapsto x$  satisfies

$$f \circ g = \text{id}_p \text{ and } g \circ f = C_x.$$

By c),  $C_x \simeq \text{id}_X$ , so  $g \circ f \simeq \text{id}_X$ . So proved  $\square$

Corollary. Contractible space is Path-Connected.

Proof. Suppose that  $X$  is Contractible space. Let  $x, y \in X$  be given,

Then,  $C_x \simeq \text{id}_X \simeq C_y$ , so there exists a Homotopy

$$F: X \times I \rightarrow X \text{ such that } \begin{cases} F(z, 0) = C_x(z) = x \text{ for all } z \in X \\ F(z, 1) = C_y(z) = y \text{ for all } z \in X. \end{cases}$$

Now,  $p: I \rightarrow X: t \mapsto F(x, t)$  is a path from  $x$  to  $y$ .  $\square$

Basic Propositions.

Prop. Let  $f, g: X \rightarrow Y$  be continuous and  $f \simeq g$ .

If  $f$  is Homotopy equivalence, then  $g$  is also Homotopy equivalence

Proof. Let  $h: Y \rightarrow X$  be the Homotopy inverse of  $f$ .

$$h \circ f \simeq \text{id}_X \text{ and } f \circ h \simeq \text{id}_Y.$$

Meanwhile, let  $F: X \times I \rightarrow Y$  be the Homotopy from  $f$  to  $g$ .

Then,  $H: X \times I \rightarrow X: (x, t) \mapsto h(F(x, t))$  satisfying  $\begin{cases} H(x, 0) = h(f(x)) \\ H(x, 1) = h(g(x)) \end{cases}$

So,  $h \circ f \simeq h \circ g$ , thus  $h \circ g \simeq \text{id}_X$ . And, define

$$G: Y \times I \rightarrow Y: (y, t) \mapsto F(h(y), t) \text{ satisfying } \begin{cases} G(y, 0) = f(h(y)) \\ G(y, 1) = g(h(y)) \end{cases}$$

So,  $f \circ h \simeq g \circ h$ , thus  $g \circ h \simeq \text{id}_Y$ . So proved.  $\square$

Prop. Let  $f, g: X \rightarrow \prod_{\alpha \in \Lambda} Y_\alpha$  be continuous maps. Then, for projection  $\pi_\alpha$ ,

$f \simeq g$  if and only if for all  $\alpha \in \Lambda$ ,  $\pi_\alpha \circ f \simeq \pi_\alpha \circ g$ .

Proof. Suppose that there is a Homotopy  $F: X \times I \rightarrow \prod_{\alpha \in \Lambda} Y_\alpha$ . (That is,  $f \simeq g$ ).

Then, the projection  $\pi_\beta: \prod_{\alpha \in \Lambda} Y_\alpha$  induces a Homotopy

$$(\pi_\beta \circ F): X \times I \rightarrow Y_\beta: (x, t) \mapsto \pi_\beta(F(x, t)).$$

Satisfying  $(\pi_\beta \circ F)(x, 0) = (\pi_\beta \circ f)(x)$  and  $(\pi_\beta \circ F)(x, 1) = (\pi_\beta \circ g)(x)$ , so that  $\pi_\beta \circ f \simeq \pi_\beta \circ g$ . Conversely, suppose that for all  $\alpha \in \Lambda$ ,  $\pi_\alpha \circ f \simeq \pi_\alpha \circ g$ .

For each  $\alpha \in \Lambda$ , let  $F_\alpha: X \times I \rightarrow Y_\alpha$  be the Homotopy from  $\pi_\alpha \circ f$  to  $\pi_\alpha \circ g$ .

Define  $F: X \times I \rightarrow \prod_{\alpha \in \Lambda} Y_\alpha: (x, t) \mapsto (F_\alpha(x, t))_{\alpha \in \Lambda}$

Then, by the property of product space,  $F$  is continuous, so, it is a Homotopy from  $f$  to  $g$ .

Prop. If  $X_\alpha \simeq Y_\alpha$  for all  $\alpha \in \Lambda$ , then  $\prod_{\alpha \in \Lambda} X_\alpha \simeq \prod_{\alpha \in \Lambda} Y_\alpha$ .

$\square$

Prop. If  $Y$  is Contractible, then  $X \times Y \cong X$ .

Proof. Let  $P = \{p\}$  be a one-point set space which is Homotopy equivalent to  $Y$ . That is, there exist two Continuous maps  $f: P \rightarrow Y$  and  $g: Y \rightarrow P$  satisfy

$$f \circ g \cong \text{id}_Y \quad \text{and} \quad g \circ f \cong \text{id}_P.$$

Actually,  $g \circ f$  must equals to  $\text{id}_P$ . To show  $X \times Y \cong X \times P$ , define the Homotopy equivalence

$$(\text{id}_X \times g): X \times Y \rightarrow X \times P; (x, y) \mapsto (x, p)$$

with the Homotopy inverse

$$(\text{id}_X \times f): X \times P \rightarrow X \times Y; (x, p) \mapsto (x, f(p)).$$

Then,  $(\text{id}_X \times g) \circ (\text{id}_X \times f) \equiv \text{id}_X \times \text{id}_P = \text{id}_{X \times P}$ .

Meanwhile, to show  $(\text{id}_X \times f) \circ (\text{id}_X \times g) \cong \text{id}_{X \times Y}$ , let

$$F: Y \times I \rightarrow Y \text{ such that } F(y, 0) = (f \circ g)(y) \text{ and } F(y, 1) \equiv \text{id}_Y.$$

be the Homotopy. Now, the map

$$H: (X \times Y) \times I \rightarrow X \times Y; ((x, y), t) \mapsto (x, F(y, t))$$

Finally,  $H(\cdot, 0) = (x, (f \circ g)(y))$  and  $H(\cdot, 1) = (x, \text{id}_Y(y)) = (x, y)$ ,

so  $(\text{id}_X \times f) \circ (\text{id}_X \times g) \cong \text{id}_{X \times Y}$ . Consequently,  $X \times Y \cong X \times \{p\}$  and since  $X \cong X \times \{p\}$  thus  $X \times Y \cong X \times \{p\} \cong X$ .  $\square$

Prop. Let  $X$  be any space, and  $Y$  be a contractible.  
Then, for any continuous map  $f, g: X \rightarrow Y$ ,  $f \simeq g$ .  
Moreover, every continuous map  $h: X \rightarrow Y$  is null-homotopic.

Prop. Let  $X$  be a Topological space, and let

$$CX = X \times I / X \times \{1\}$$

be a Cone, where  $I = [0, 1]$ .

Then,  $CX$  is Contractible. Moreover, for any space  $Y$  and continuous  $f: X \rightarrow Y$ ,  $f: X \rightarrow Y$  is null-homotopic if and only if there is an extension  $\tilde{f}: CX \rightarrow Y$ .

Proof. Let  $P = \{P_0\}$  for fixed  $P_0 = [(x_0, 1)] \in CX$ .

$f: CX \rightarrow P: x \mapsto P_0$  is a Constant map, so continuous with homotopy inverse  $g: P \rightarrow CX: P_0 \mapsto P_0$  so that  $f \circ g = \text{id}_P \simeq \text{id}_P$ .

And,  $g \circ f(x) = P_0$  is a Constant map, let's show  $g \circ f \simeq \text{id}_{CX}$ .

Construct a Homotopy between  $g \circ f$  and  $\text{id}_{CX}$ . Define

$$H: CX \times I \rightarrow CX; [(x, s), t] \mapsto [(x, (1-t) \cdot s + t)]$$

Then,  $H([(x, s)], 0) = [(x, s)]$  and  $H([(x, s)], 1) = [(x, 1)]$

(Specifically,  $H$  is well-defined Continuous map,

(Note:  $H([(x, s)], 0) = [(x, s)]$ )

